

Bounded Stepanov almost automorphy is invariant under every finite variable exponent

Candidate full solution to the problem in arXiv:1903.05869

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Status. Candidate full solution, pending expert review.

Source. T. Diagana and M. Kostić, *Almost automorphic and asymptotically almost automorphic type functions in Lebesgue spaces with variable exponents $L^{p(x)}$* , arXiv:1903.05869, unnumbered Problem on printed page 9 [1].

1 The question

For a measurable exponent $p : [0, 1] \rightarrow [1, \infty)$, write

$$p^- = \operatorname{ess\,inf} p(x), \quad p^+ = \operatorname{ess\,sup} p(x),$$

and let $D_+([0, 1])$ consist of those p with $1 \leq p^- \leq p(x) \leq p^+ < \infty$ almost everywhere. The source uses $AAS^{p(x)}(\mathbb{R} : X)$ for Stepanov $p(x)$ -almost automorphic functions and $AAAS^{p(x)}([0, \infty) : X)$ for their asymptotic version.

S^1 -almost automorphic and $\hat{q} \in C_0([0, \infty) : L^1([0, 1] : X))$ because $L^{p(x)}([0, 1] : X) \hookrightarrow L^1([0, 1] : X)$; therefore, $f \in AAAS^1([0, \infty) : X)$. Using the fact that $L_S^{p(x)}([0, \infty) : X) \hookrightarrow L_S^1([0, \infty) : X)$, it readily follows that there is a finite constant $c > 0$, independent of $f(\cdot)$, such that

$$\|f\|_{L_S^{p(x)}([0, \infty) : X)} \leq c \|f\|_{L_S^1([0, \infty) : X)}.$$

This completes the proof. $\square$

PROBLEM. In [4], we have proved the following: If $p \in D_+([0, 1])$, then

$$L^\infty(\mathbb{R} : X) \cap APS^{p(x)}(\mathbb{R} : X) = L^\infty(\mathbb{R} : X) \cap APS^1(\mathbb{R} : X)$$

and

$$L^\infty([0, \infty) : X) \cap AAPS^{p(x)}([0, \infty) : X) = L^\infty([0, \infty) : X) \cap AAPS^1([0, \infty) : X).$$

The proof given in the above-mentioned paper does not work for almost automorphy with variable exponent. Because of that, we would like to ask whether the assumption $p \in D_+([0, 1])$ implies

$$L^\infty(\mathbb{R} : X) \cap AAS^{p(x)}(\mathbb{R} : X) = L^\infty(\mathbb{R} : X) \cap AAS^1(\mathbb{R} : X)$$

and

$$L^\infty([0, \infty) : X) \cap AAAS^{p(x)}([0, \infty) : X) = L^\infty([0, \infty) : X) \cap AAAS^1([0, \infty) : X)?$$

The subsequent lemma can be deduced following the lines of proof of [4] Proposition 3.5]:

Lemma 3.8. *Assume that $v \in \mathcal{P}([0, 1])$ and $a \in C_n([0, \infty) : X)$. Then $a \in$*

Figure 1: The exact problem in the archived source PDF, printed page 9.

Transcription. Does $p \in D_+([0, 1])$ imply

$$L^\infty(\mathbb{R} : X) \cap AAS^{p(x)}(\mathbb{R} : X) = L^\infty(\mathbb{R} : X) \cap AAS^1(\mathbb{R} : X) \tag{Q1}$$

and

$$L^\infty([0, \infty) : X) \cap AAAS^{p(x)}([0, \infty) : X) = L^\infty([0, \infty) : X) \cap AAAS^1([0, \infty) : X)? \quad (\text{Q2})$$

The answer to both questions is yes.

2 Proof Intuition

The sole analytic mechanism is bounded-convergence interpolation. On a unit interval, if $u_j \rightarrow 0$ in L^1 , all u_j have one common bound, and $p^+ < \infty$, then $u_j \rightarrow 0$ in $L^{p(x)}$. The variable-exponent modular is bounded by a fixed multiple of $\|u_j\|_1$.

For (Q1), the local L^1 limit profile of translates of a bounded function inherits the same essential bound. Both defining limits upgrade from L^1 to $L^{p(x)}$.

For (Q2), a witnessing decomposition $f = g + q$ need not initially present g and q as bounded. Translate it along integers tending to infinity. The remainder disappears locally in L^1 , so the forward almost-automorphic limit of g is also a limit of translates of bounded f . The reverse limit then forces g itself to be bounded. Thus $q = f - g$ is bounded too.

3 Two lemmas

The Luxemburg norm on $L^{p(x)}([0, 1] : X)$ is

$$\|u\|_{p(x)} = \inf \left\{ \lambda > 0 : \int_0^1 \left(\frac{\|u(s)\|}{\lambda} \right)^{p(s)} ds \leq 1 \right\}.$$

Lemma 1 (bounded L^1 convergence upgrades). *Let $p \in D_+([0, 1])$. Suppose $u_j : [0, 1] \rightarrow X$ satisfy $\|u_j(s)\| \leq B$ almost everywhere and $\|u_j\|_1 \rightarrow 0$. Then $\|u_j\|_{p(x)} \rightarrow 0$.*

Proof. Fix $\varepsilon > 0$ and put $v_j = \|u_j\|/\varepsilon$. Since $1 \leq p(s) \leq p^+$ and $0 \leq v_j \leq B/\varepsilon$,

$$v_j^{p(s)} \leq v_j + v_j^{p^+} \leq \left[1 + \left(\frac{B}{\varepsilon} \right)^{p^+-1} \right] v_j.$$

The modular of u_j/ε therefore tends to zero. It is at most one for all large j , so $\|u_j\|_{p(x)} \leq \varepsilon$. $\square$

Lemma 2 (bounded asymptotic data have bounded recurrent part). *Suppose $f \in L^\infty([0, \infty) : X)$ and*

$$f = g + q, \quad g \in AAS^1(\mathbb{R} : X), \quad \|\widehat{q}(t)\|_{L^1([0, 1] : X)} \rightarrow 0, \quad (1)$$

where $\widehat{q}(t)(s) = q(t + s)$. Then

$$\|g\|_\infty \leq \|f\|_\infty, \quad \|q\|_\infty \leq 2\|f\|_\infty. \quad (2)$$

Proof. Let $M = \|f\|_\infty$. Apply Stepanov 1-almost automorphy of g to the integer sequence tending to infinity. Passing to a subsequence $a_j \rightarrow \infty$, there is a profile h such that, for every $t \in \mathbb{R}$,

$$\|g(a_j + t + \cdot) - h(t + \cdot)\|_1 \rightarrow 0, \quad \|h(t - a_j + \cdot) - g(t + \cdot)\|_1 \rightarrow 0. \quad (3)$$

For each integer m , (1) and $f = g + q$ show that $f(a_j + m + \cdot) \rightarrow h(m + \cdot)$ in $L^1([0, 1] : X)$. Every function on the left is bounded by M , so its L^1 limit is bounded by M almost everywhere. Taking all integer m proves $\|h\|_\infty \leq M$. The reverse limit in (3), again on all integer unit intervals, gives $\|g\|_\infty \leq M$. Finally $q = f - g$ on the half-line, proving (2). $\square$

4 The affirmative answer

Theorem 1. *For every complex Banach space X and every $p \in D_+([0, 1])$, both equalities (Q1) and (Q2) hold.*

Proof. The source proves the inclusions from the $p(x)$ classes into the exponent-one classes [1, Theorem 3.7(i)]; they also follow from the finite-measure embedding $L^{p(x)}([0, 1] : X) \hookrightarrow L^1([0, 1] : X)$.

Let $f \in L^\infty(\mathbb{R} : X) \cap AAS^1(\mathbb{R} : X)$ and set $M = \|f\|_\infty$. Given any real sequence, choose a subsequence a_j and a profile g which realize the two local L^1 limits defining AAS^1 . On each integer unit interval, g is an L^1 limit of translates of f , hence $\|g\|_\infty \leq M$. For each fixed t , both differences

$$f(a_j + t + \cdot) - g(t + \cdot), \quad g(t - a_j + \cdot) - f(t + \cdot)$$

are bounded by $2M$ and tend to zero in $L^1([0, 1] : X)$. Lemma 1 upgrades both limits to $L^{p(x)}$. Bounded g is Stepanov $p(x)$ -bounded, so $f \in AAS^{p(x)}(\mathbb{R} : X)$. This proves (Q1).

Now let $f \in L^\infty([0, \infty) : X) \cap AAAS^1([0, \infty) : X)$ and choose a decomposition (1). Lemma 2 makes both g and q bounded. The just-proved result gives $g \in AAS^{p(x)}(\mathbb{R} : X)$. Since q is bounded and $\|\widehat{q}(t)\|_1 \rightarrow 0$, Lemma 1 gives

$$\|\widehat{q}(t)\|_{L^{p(x)}([0, 1] : X)} \rightarrow 0.$$

Boundedness also makes q Stepanov $p(x)$ -bounded. The same decomposition therefore witnesses

$$f \in AAAS^{p(x)}([0, \infty) : X),$$

which proves (Q2). □

5 Verification, limitations, and novelty

The script `code/verify_modular_upgrade.py` checks the modular inequality in Lemma 1 on 1,200 randomized finite grids, with variable exponents between 1 and 9. Every check passes. This is a normalization check only; the proof is analytic and exact.

The finite upper exponent $p^+ < \infty$ is essential to this proof. The source notes different behavior when $p(x) = \infty$ on a nontrivial set. All equalities are understood modulo almost-everywhere equality.

Novelty check. On 12 August 2026, the run indexes and bounded arXiv/web searches used the exact source title, the two displayed class equalities, and phrases involving bounded Stepanov $p(x)$ -almost automorphy and D_+ exponent invariance. Searches found the source, later surveys and monographs, and related variable-exponent Stepanov work, but no statement of these equalities or explicit answer to the source problem. Novelty confidence is moderate.

Human-review recommendation. Review is recommended. The decisive points are the modular estimate in Lemma 1 and the use of both directions of Stepanov almost automorphy in Lemma 2; no deep external theorem is imported.

References

- [1] T. Diagana and M. Kostić, *Almost automorphic and asymptotically almost automorphic type functions in Lebesgue spaces with variable exponents $L^{p(x)}$* , arXiv:1903.05869 (2019).